

Delhi Public School

Subject: Biology | Grade: 12th | Class: 12th

Time: 2 Hours

Maximum Marks: 13

Name: _____

Roll No: _____

Section: _____

General Instructions:

1. Read all questions carefully before answering.
2. Write neatly and legibly.

Section A

Objective Type Questions. Choose the correct option for each of the following questions. Each question carries 1 mark. Time recommended: 15 minutes.

Q1. Which of the following is a classic example of codominance in humans? [Easy] [1]

- a. Skin color
- b. ABO blood grouping
- c. Height
- d. Eye color

Q2. In a cross between red-flowered (RR) and white-flowered (rr) *Mirabilis jalapa* plants, the F1 plants are all pink (Rr). When F1 plants are self-pollinated, what will be the phenotypic ratio in the F2 generation? [1]

[Moderate]

- a. 3 Red : 1 White
- b. 1 Red : 2 Pink : 1 White
- c. 9 Red : 3 Pink : 4 White
- d. All Pink

Q3. In a stable population at Hardy-Weinberg equilibrium, the frequency of a recessive allele (q) is 0.4. What is the frequency of the heterozygous carrier genotype (2pq) in this population? [1]

[Moderate]

- a. 0.16
- b. 0.36
- c. 0.48
- d. 0.24

Q4. The presence of homologous organs in different organisms, such as the forelimbs of whales, bats, and humans, indicates which type of evolution? [1]

[Easy]

- a. Convergent evolution
- b. Divergent evolution
- c. Saltation
- d. Co-evolution

Q5. What is the expected phenotypic ratio in the F2 generation of a Mendelian dihybrid cross involving seed shape and seed color? [1]

[Moderate]

- a. 3:1
- b. 1:2:1
- c. 9:3:3:1
- d. 1:1:1:1

Section B

Short Answer Questions. Answer the following questions in 30-50 words. Each question carries 2 marks.

Q6. What is a test cross? Explain its significance in determining the genotype of an organism showing a dominant phenotype. [2]

[Easy]

Q7. Differentiate between incomplete dominance and codominance, providing one biological example for each. [Moderate] **[2]**

Q8. State the Hardy-Weinberg principle. List any three factors that are known to disrupt Hardy-Weinberg equilibrium in a natural population. [Hard] **[2]**

Section C

Long Answer Questions. Answer the following detailed questions. Each question carries 1 mark. Time recommended: 20 minutes.

Q9. Using the phenomenon of Industrial Melanism observed in England (moths before and after industrialization), explain Charles Darwin's theory of Natural Selection. How does this serve as an observable example of evolutionary change? [Moderate] **[1]**

Q10. State Mendel's Law of Independent Assortment. Demonstrate this law using a dihybrid cross between a homozygous round yellow-seeded pea plant (RRYY) and a wrinkled green-seeded pea plant (rryy). Show the gametes, F1 generation, and the F2 generation punnett square representation. [Hard] **[1]**

— End of Question Paper —